Machine Learning 1

Homework 1

1 problem 1

1(a). Since $\mathbb{E}_{X\sim P}[U(a,X)] = (d+\frac{2}{3}a)*0.85+(d-a)*0.15 = d+\frac{5}{12}a$ and $0 \le a \le d$, we have $a^* = d$. If it takes variance/risk of bet into account it would make more sense.

1(b). Since $\mathbb{E}_{X\sim P}[U(a,X)] = \log(d+\frac{2}{3}a)*0.85+\log(d-a)*0.15$,

$$f(a) = \log(d+\frac{2}{3}a)*0.85+\log(d-a)*0.15, f'(a) = \frac{17}{30}/(\frac{2}{3}a+d) - 0.15/(d-a)$$

solve $f'(a) = 0$ we have $a^* = 0.625d$, and furtheremore $f''(a^*) < 0$.

1(c). No. I am personally more risk-averse.

2 problem 2

Note that

$$\begin{aligned}\arg\min \mathbb{E}_{X,Y}[L(Y,f(X))] &= \arg\min \mathbb{E}_{X,Y}[-Y\log f(X) - (1-Y)\log(1-f(X))] \\ &= \arg\min \mathbb{E}_X[\mathbb{E}_{Y|X}[-Y\log f(X) - (1-Y)\log(1-f(X))|X]] \\ &= \arg\min \mathbb{E}_X[-p(Y=1|X)\log f(X) - (1-p(Y=1|X=x))\log(1-f(X))]\end{aligned}$$

take derivative with respect to $f(x)$ gives $f^*(x) = p(Y=1|X=x)$ and furthermore the second derivative is negative, which concludes the result.

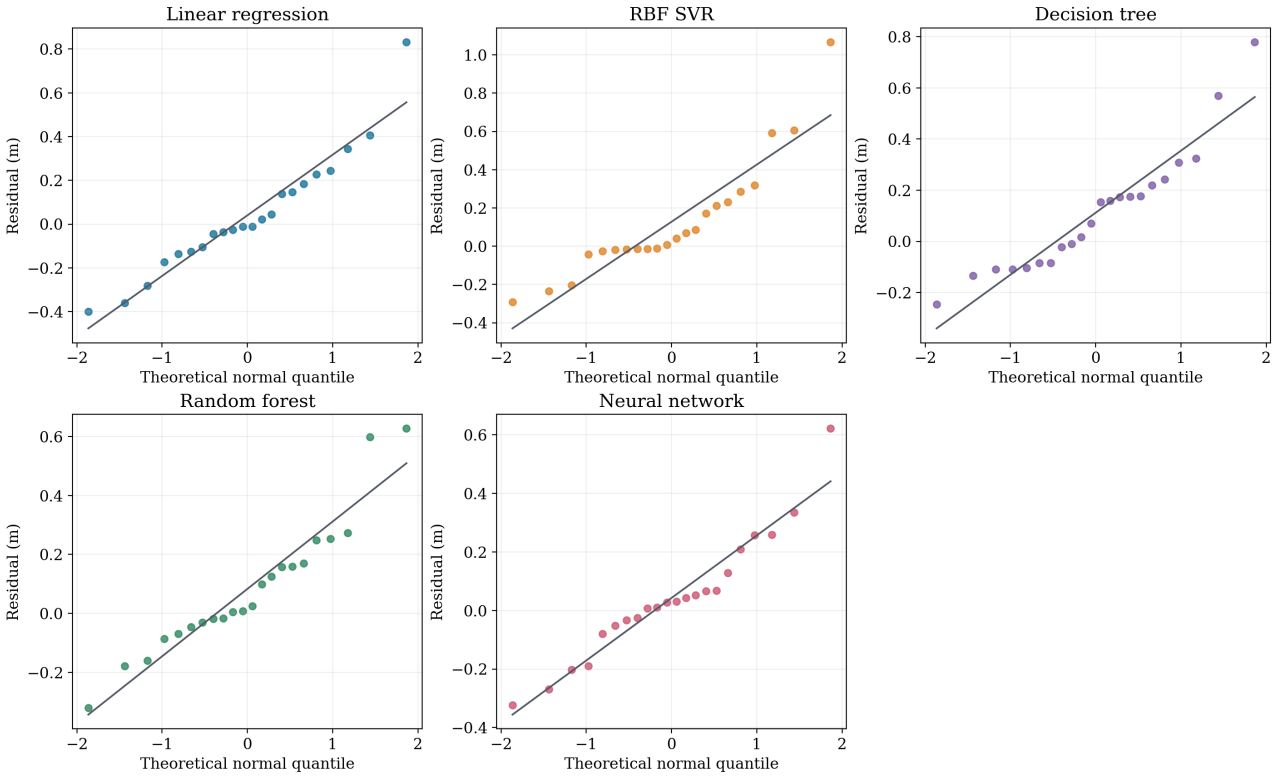
3 problem 3

One of the reason that there is already bias comes from the finite sample compared to ground truth distribution, so in such case we only need to achieve a loss with same order as sampling bias instead of a zero loss.

4 problem 4

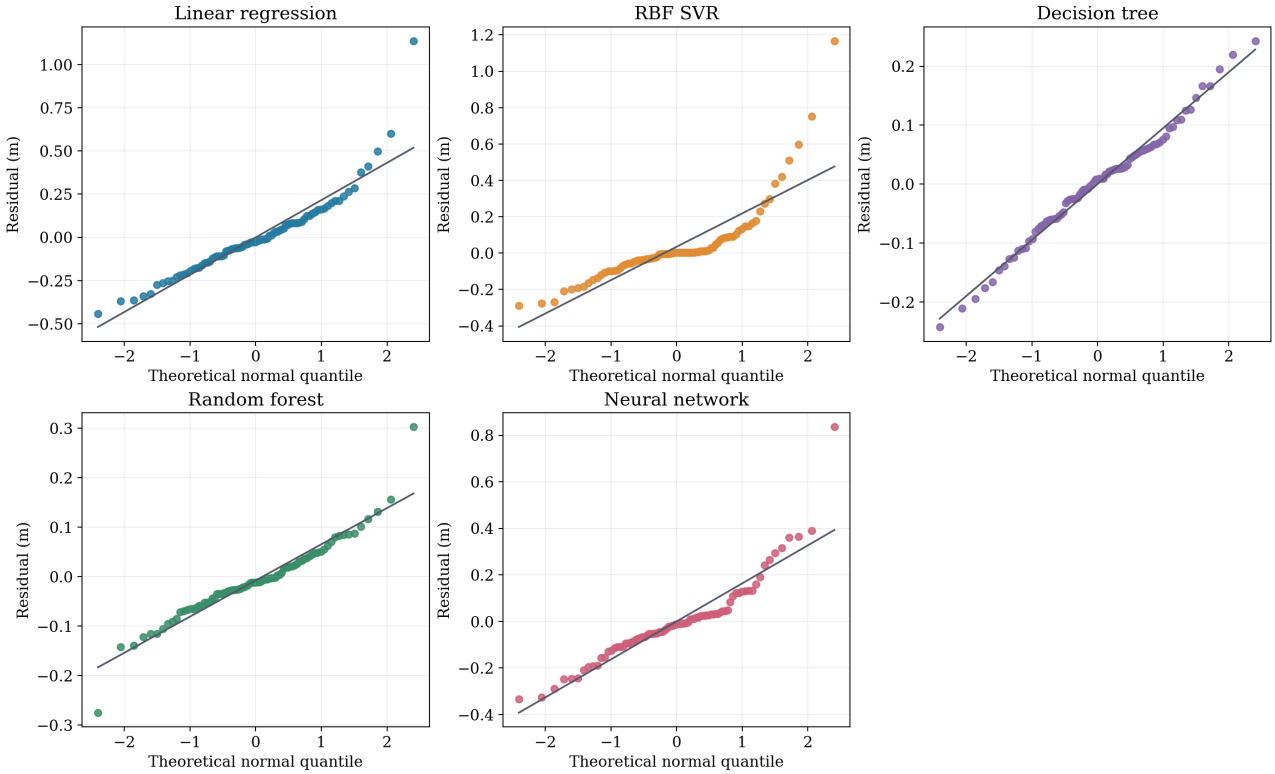
Lets just focus on the fitted residual on train/test data. We use 5 different models including linear regrassion, support vector regrassion, decision tree, random forest and neural network. We present qq plot and a normal test of five models.

Test residuals ($n = 22$)**.**



Model	`normaltest` K^2	p-value
Linear regression	7.2997	0.025995
RBF SVR	13.0913	0.001436
Decision tree	7.0637	0.029250
Random forest	5.1604	0.075758
Neural network	4.8691	0.087637

Training residuals ($n = 85$).



Model	`normaltest` K^2	p-value
Linear regression	46.5802	7.678×10^{-11}
RBF SVR	71.4519	3.051×10^{-16}
Decision tree	0.8014	0.669853
Random forest	16.2658	2.937×10^{-4}
Neural network	42.2184	6.798×10^{-10}

On test set, for linear regression, SVR and dicision tree, we reject the hepothesis of a normal residual under significance level 0.05. The qq plot also exhibits a deviation on the tail. On train set, we reject all the model except decision tree. However the decision tree has a strong overfitting on train set.

5 problem 5

Both datasets use an 80/20 split with seed 42. We compare 200 and 500 epochs along the same SGD trajectory with mini-batch shuffle seed 42. Each model uses its full training set with fixed hyperparameters, without cross-validation. Features are standardized using training data only. Classification probabilities are clipped to $[10^{-6}, 1-10^{-6}]$ before BCE; gradient norms are clipped at 5. The learning rate is $\eta_e = \eta_0/(1+e/100)$, where e starts at zero.

Classification ($n_{\text{train}} = 455$, $n_{\text{test}} = 114$).

Model	Epochs	Correct	Test accuracy	Δ accuracy
sklearn logistic (L2)	–	112/114	0.9825	+0.0000
PyTorch logistic (SGD)	200	110/114	0.9649	-0.0175
PyTorch logistic (SGD)	500	110/114	0.9649	-0.0175
PyTorch probit (SGD)	200	109/114	0.9561	-0.0263
PyTorch probit (SGD)	500	110/114	0.9649	-0.0175

For the additional test metrics, malignant ($y = 0$) is the positive class. ROC-AUC uses its predicted probability; F_1, sensitivity, and specificity use the same 0.5 classification threshold. Higher values are better.

Model	Epochs	ROC-AUC	F_1	Sensitivity	Specificity
sklearn logistic (L2)	–	0.9954	0.9762	0.9762	0.9861
PyTorch logistic (SGD)	200	0.9954	0.9535	0.9762	0.9583
PyTorch logistic (SGD)	500	0.9957	0.9535	0.9762	0.9583
PyTorch probit (SGD)	200	0.9954	0.9425	0.9762	0.9444
PyTorch probit (SGD)	500	0.9954	0.9535	0.9762	0.9583

Classification (a). SGD uses batch size 64, initial learning rate 0.05, and momentum 0. The sigmoid model achieves 96.49% accuracy at both 200 and 500 epochs, 1.75 percentage points below the baseline. ROC-AUC is almost identical to the baseline. Sensitivity is also identical, while specificity is lower: the model produces three false positives, compared with one for sklearn. The PyTorch model minimizes BCE without a penalty; sklearn uses L2 regularization.

Classification (b). With the same SGD settings, replacing sigmoid with the standard normal CDF improves accuracy from 95.61% at 200 epochs to 96.49% at 500 epochs. False positives decrease from four to three; sensitivity is unchanged. The 500-epoch result remains 1.75 percentage points below the baseline, with almost identical ROC-AUC. Both models use clipped BCE.

Regression ($n_{\text{train}} = 85$, $n_{\text{test}} = 22$).

Model	Epochs	Test MAE (m)	Δ MAE (m)
sklearn linear	–	0.195393	+0.000000
PyTorch linear (MSE, SGD)	200	0.195232	-0.000161
PyTorch linear (MSE, SGD)	500	0.195753	+0.000360
PyTorch linear (t_5 NLL, SGD)	200	0.194035	-0.001358
PyTorch linear (t_5 NLL, SGD)	500	0.194308	-0.001086

RMSE gives greater weight to large errors than MAE; lower is better. Higher R^2 indicates better fit relative to a constant predictor using the test-set mean.

Model	Epochs	Test RMSE (m)	Test R^2
sklearn linear	–	0.270543	0.629198
PyTorch linear (MSE, SGD)	200	0.270485	0.629356
PyTorch linear (MSE, SGD)	500	0.269541	0.631939
PyTorch linear (t_5 NLL, SGD)	200	0.272995	0.622445
PyTorch linear (t_5 NLL, SGD)	500	0.272095	0.624931

Regression (a). SGD uses batch size 32, initial learning rate 0.03, and momentum 0. With MSE, test MAE is 0.195232 m at 200 epochs and 0.195753 m at 500 epochs, compared with 0.195393 m for the baseline. The 500-epoch model has slightly lower RMSE and higher R^2 than the baseline, despite its slightly higher MAE. These small differences are consistent with finite stochastic updates near the least-squares solution.

Regression (b). With the same SGD settings, minimizing the Student-t negative log-likelihood with $\nu = 5$ and fixed scale 1 gives test MAE 0.194035 m at 200 epochs and 0.194308 m at 500 epochs. Both are lower than the baseline, but their RMSE is higher and R^2 is lower. Thus, the improvement depends on the evaluation metric. ALE remains in its original units.